

RepairAgent: An Autonomous, LLM-Based Agent for Program Repair

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Automated program repair has emerged as a powerful technique to mitigate the impact of software bugs on system reliability and user experience. This paper introduces RepairAgent, the first work to address the program repair challenge through an autonomous agent based on a large language model (LLM). Unlike existing deep learning-based approaches, which prompt a model with a fixed prompt or in a fixed feedback loop, our work treats the LLM as an agent capable of autonomously planning and executing actions to fix bugs by invoking suitable tools. Repair Agent freely interleaves gathering information about the bug, gathering repair ingredients, and validating fixes, while deciding which tools to invoke based on the gathered information and feedback from previous fix attempts. Key contributions that enable RepairAgent include a set of tools that are useful for program repair, a dynamically updated prompt format that allows the LLM to interact with these tools, and a finite state machine that guides the agent in invoking the tools. Our evaluation on the popular Defects4J dataset demonstrates RepairAgent’s effectiveness in autonomously repairing 164 bugs, including 39 bugs not fixed by prior techniques. Interacting with the LLM imposes an average cost of 270k tokens per bug, which, under the current pricing of OpenAI’s GPT-3.5 model, translates to 14 cents per bug. To the best of our knowledge, this work is the first to present an autonomous, LLM-based agent for program repair, paving the way for future agent-based techniques in software engineering.

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1 Introduction

Software bugs can lead to system failures, security vulnerabilities, and compromised user experience. Manually fixing bugs demands significant time and user effort, but **Automated Program Repair (APR)** can significantly reduce the effort via automatic bug resolution. Various approaches have been explored in this regard [Le Goues et al. 2019], incl. techniques based on manually designed [Le Goues et al. 2012], [Liu et al. 2019] and (semi-)automatically extracted [Kim et al. 2013], [Bader et al. 2019] fix patterns, based on symbolic constraints [Nguyen et al. 2013], [Xuan et al. 2016], and various machine learning based approaches [Long and Rinard 2016], [Gupta et al. 2017], [Tufano et al. 2019].

The current state-of-the-art in APR is predominated by LLMs. In the first generation of LLM based APR, the model expects a prompt with the buggy code and produces the fixed version [Xia et al. 2023], [Jiang et al. 2023], but further development introduces an iterative approach: the LLM is queried repeatedly based on feedback obtained from previous fix attempts [Xia and Zhang 2023], [Kang et al. 2023a], [Ye and Monperrus 2024].

But this approach has a few limitations since the hard-coded feedback loops in the LLMs do not allow them to gather information about existing bugs or existing code that may provide them with debugging ingredients, unlike manual

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human debugging which typically incorporates a temporal interleaving of gathering information to gain understanding of the bug, searching code that might help in debugging and then attempting various versions of fixes [Ko et al. 2006], [Böhme et al. 2017].

This paper represents **RepairAgent**, the first autonomous, LLM based agent for automated program repair. The LLM is treated as an autonomous agent capable of planning and executing actions to fix a bug. The LLM is equipped with a set of repair-specific tools. How and when to use these tools aren't hard coded, rather the LLM gets to autonomously decide.

The approach is enabled via three key components: a general purpose LLM e.g. GPT-3.5, a set of tools for the LLM to use (in this approach, 14 tools in total have been used), and a middleware to establish communication between the tools and the LLM.

The novelty of the approach was tested using all 835 bugs of the Defects4J [Just et al. 2014] dataset. 164 bugs were successfully fixed, including 39 bugs that were not fixed by prior state-of-the-art work [Xia and Zhang 2023] [Ye and Monperrus 2024].

The interaction with the LLM imposes a cost of $\sim 270K$ tokens per bug on average, which translates to about **14 cents** per bug.

In summary, this paper contributes the following:

- An autonomous, LLM-based agent for program repair.
- A dynamically updated prompt format that guides the LLM through the bug repair process.
- A set of tools that enable a LLM to perform steps a human developer would take when fixing a bug.
- A middleware that guides via a finite state machine how the LLM interacts with the tools.
- Empirical evidence that RepairAgent establishes a new state of the art in program repair.
- We release our implementation as open-source: <https://github.com/sola-st/RepairAgent>.

To the best of our knowledge, this is the first work to present an autonomous, LLM-based agent for program repair, paving the way for future agent-based techniques in software engineering.

2 Background (LLM-Based Autonomous Agents)

Because of extensive training, LLMs have demonstrated remarkable abilities in achieving human-level performance in a variety of tasks [Bubeck et al. 2023]. A promising way to use these LLMs are LLM-based agents, which mean LLM-based techniques with two properties: (1) The LLM can autonomously plan for and execute a set of actions to achieve a particular goal, unlike hard-coded algorithmic processes, and (2) The actions by the LLM can invoke external tools, enabling it to interact with its environment [Schick et al. 2023] [Patil et al. 2023]. In the context of Software Engineering, the tools could be tools usually used by developers, e.g., as part of an integrated development environment (IDE). Basically, we query the LLM with the current state of the world, the goal and a set of actions to perform. The feedback is integrated into the next prompt.

3 Method

3.1 Overview

Fig. 1 provides an overview of RepairAgent.

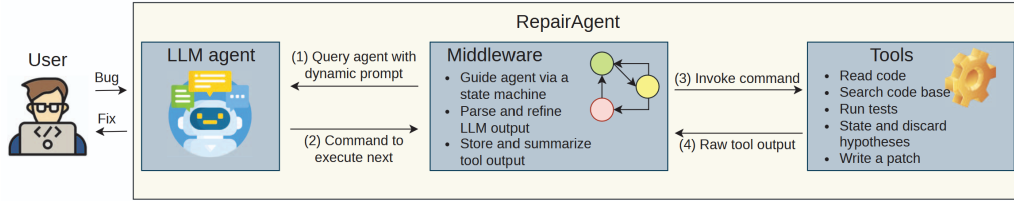


Fig. 1. Overview of RepairAgent.

3.2 Terminology

RepairAgent proceeds in multiple cycles:

Definition 1 (Cycle). A *cycle* represents one round of interaction with the LLM agent, which consists of the following steps:

- (1) Query the agent
- (2) Post-process the response
- (3) Execute the command suggested by the agent
- (4) Update the dynamic prompt based on the command's output

In each cycle, the approach queries the LLM once. The input to the model is updated based on commands invoked by the LLM, and their results, in previous cycles. We call the model input a dynamic prompt:

Definition 2 (Dynamic prompt). The *dynamic prompt* is a sequence of text sections $P = [s_0, s_1, \dots, s_n]$, where each section s_i is one of the following (where $s_i(c)$ refers to a section during a cycle c):

- A *static section*, which remains the same across all cycles, i.e., $s_i(c) = s_i(c')$ for all c, c' .
- A *dynamic section*, which may differ across cycles, i.e., there may exist c, c' with $s_i(c) \neq s_i(c')$.

3.3 Dynamic Prompting of the RepairAgent

As listed in Table 1, the prompt contains multiple static and dynamic sections.

Table 1. Sections of the dynamically updated prompt within RepairAgent

Prompt Section	Nature
Role	Static
Goals	Static
Guidelines	Static
State description	Dynamic
Available tools	Dynamic
Gathered information	Dynamic
Specification of output format	Static
Last executed command and result	Dynamic

(1) **Role**: provides the agent's area of expertise and outlines the agent's primary objective: finding and fixing bugs

(2) **Goals**: There are 5 goals for the agent to pursue:

- a) *Locate the bug* using testing and fault localization techniques

- b) *Gather information about the bug* by analyzing the relevant portion(s) of the codebase
- c) *Suggest simple fixes to the bug*
- d) *Suggest complex fixes*
- e) *Iterate over the previous goals*

(3) Guidelines: The LLM is provided with a set of guidelines. It is told that there are various kinds of bugs ranging from one-line-fix bugs to multi-line bugs. A list of recurring fix patterns is also provided that help solve many common bugs [Karampatsis and Sutton 2020] which is based on single-statement bugs in **Java**. The model is also instructed to add comments along with edited code which helps the model explain its reasoning and improve itself [Wei et al. 2022]. The model is also instructed to define a next step that can be converted into a call to a tool.

(4) State description: To guide the LLM toward using the tools effectively, a state machine is defined that constrains which tool(s) can be used at any given time. This is to stop the LLM to get lost in aimless exploration, which has happened very frequently as per the results of earlier tests. Fig. 2 shows the finite state machine. The set of tools are described in Section 3.4. The agent is free to move between any state via tools.

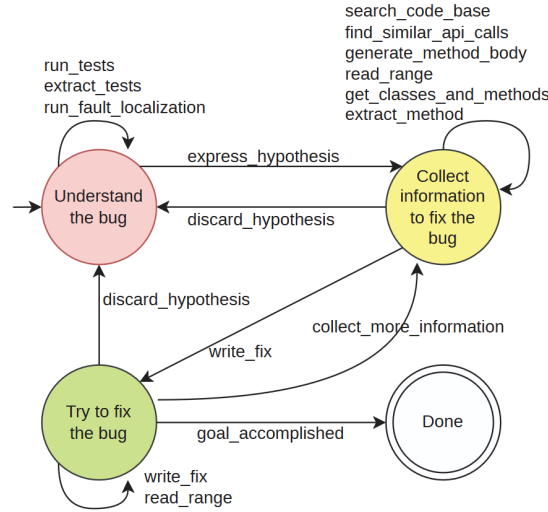


Fig. 2. State machine to guide selection of tools.

- *Understand the bugs:* The agent can collect information and bug location from testing results. It can then formulate a hypothesis based on its knowledge.
- *Collect information to fix the bug:* The agent can search for specific repair methods or read relevant code. After gathering enough information, it moves on to the next state.
- *Try to fix the bug:* The agent can try to fix the bug based on the hypothesis and information gathered. If necessary, the agent can go back and establish a new hypothesis or gather additional information.

(5) Available tools: This section describes the currently available tools at any state.

(6) Gathered information: This serves as a memory for the agent, enabling it to recall information from previous cycles. This section is divided into various subsections that contain outputs from different tools.

(7) **Specification of output format:** The agent gives one response per cycle in the format specified by Fig. 3. The "thoughts" section is the agent's reasoning when deciding what to do for the next command.

```
interface Response {
  thoughts: string;
  command: {
    name: string;
    args: Record<string, any>;
  };
};
```

Fig. 3. Output format specification.

```
{
  "thoughts": "With the hypothesis in mind,
it's time to gather more information to
formulate potential fixes for the bug. I
should focus on understanding the context
around the condition
'if (x < 0 && prev == '-')'
to come up with effective solutions.",
  "command": {
    "name": "search_code_base",
    "args": {
      "key_words":
        [ "addNumber",
          "CodeConsumer",
          "if (x < 0 && prev == '-')" ]
    }
  }
}
```

Fig. 4. Example agent response layout.

(8) **Last executed command and result:** This section helps the agent remind the last steps taken in case anything went wrong during execution. It also helps the agent remember how many cycles have been executed and how many are left.

3.4 Tools for the Agent to Use

A key novelty of the approach is letting the LLM agent autonomously decide which tools to invoke to fix a bug. The tools provided to the agent (Table 2) are inspired by the tools developers typically use in their IDEs.

- a) *Reading and Extracting Code:* Instead of hard-coding the context provided to the LLM [Xia and Zhang 2023], [Kang et al. 2023a], [Ye and Monperrus 2024], the agent gets to decide which parts of the code to read. It uses tools like `read_range` to extract specific lines for a focused view, `get_classes_and_methods` for a code structure overview, `extract_method` to retrieve method implementations, and `extract_tests` to get the code of failing test cases to understand expected outputs.
- b) *Search and Generate Code:* Because human developers commonly search for code [De Grazia and Pradel 2023], the agent is equipped with tools to search for specific code snippets to gather repair ingredients. The `search_code_base` tool locates keyword instances across the entire codebase using approximate matching (splitting keywords into subtokens). The `find_similar_api_calls` tool extracts method usages to prevent the LLM from hallucinating non-existent method calls [Eghbali and Pradel 2024]. Additionally, a `generate_method_body` tool can generate new code by invoking another LLM, inspired by the success of code completion tools like Copilot [Chen et al. 2021a].
- c) *Testing and Patching:* `run_tests` executes the project's test suite and cleans the output to avoid confusing the model with irrelevant stack-trace information. `run_fault_localization` retrieves fault-localization data, either assuming perfect localization or using tools like GZoltar [Campos et al. 2012]. `write_fix` applies a JSON patch and runs the test suite; if tests fail, it reverts the changes to give the agent a clean slate. The tool also samples multiple variants (30 by default) of a suggested fix to account for almost-correct attempts.

Table 2. Repair-related tools invoked by RepairAgent.

Tool	Description
Read and extract code:	
<i>read_range</i>	Read a range of lines in a file.
<i>get_classes_and_methods</i>	Get the names of all classes and methods in a file.
<i>extract_method</i>	Given a method name, extract method implementations from a file.
<i>extract_tests</i>	Given the failure report from JUnit or ANT, extract the code of failing test cases.
Search and generate code:	
<i>search_code_base</i>	Scans all Java files within a project for a list of keywords.
<i>find_similar_api_calls</i>	Given a code snippet that calls an API, search for similar API calls in the project.
<i>generate_method_body</i>	Ask an LLM (GPT-3.5 by default) to generate the body of a method based on code proceeding the method.
Testing and patching:	
<i>run_tests</i>	Run the test suite of a project.
<i>run_fault_localization</i>	Retrieve pre-existing localization information or run a fault localization tool.
<i>write_fix</i>	Apply a patch to the code base and execute the test suite of the project. Changes are reverted automatically if tests fail. Moves the agent into the “Try to fix the bug” state.
Control:	
<i>express_hypothesis</i>	Express a hypothesis about the bug. Moves the agent into the “Collect information to fix the bug” state.
<i>collect_more_information</i>	Move the agent back to the “Collect information to fix the bug” state.
<i>discard_hypothesis</i>	Discard the current hypothesis about the bug and move back to the “Understand the bug” state.
<i>goal_accomplished</i>	Declare that the goal has been accomplished and exiting the repair process.

- d) *Control*: These tools don’t directly correspond to human developer tools, rather they allow the agent to navigate the finite state machine. *express_hypothesis* and *discard_hypothesis* enforce a structured approach to hypothesis formulation, aligning with scientific debugging [Zeller 2009], [Kang et al. 2023a]. *collect_more_information* reverts the agent to gathering mode if fixes fail, and *goal_accomplished* terminates the process once a passing fix is found.

3.5 middleware

The middleware component orchestrates the communication between the LLM agent and the tools, seamlessly executing the steps of each cycle. It handles three primary responsibilities:

- (1) **Parsing and Refining LLM Output**: At the start of a cycle, the middleware queries the LLM. Because LLMs can produce responses with syntactic errors or hallucinations (e.g., using “path” instead of “file_path”), the middleware heuristically rectifies these deviations in three steps:
 - Maps the predicted tool name to an actual tool using substring matching or a Levenshtein distance threshold (0.1 by default).
 - Maps the provided argument names using the same matching logic.
 - Heuristically corrects invalid argument values, such as replacing a broken file path with a valid one.

If these heuristics fail, or if the agent suggests the exact same command as a previous cycle, the middleware informs the LLM about the error via the “Last executed command and result” prompt section and initiates a new cycle.

- (2) **Calling the Tool:** Once a valid command is parsed, the middleware executes the corresponding tool in an isolated environment to prevent any interference with the host system or RepairAgent itself.
- (3) **Updating the Prompt:** Using the tool’s output, the middleware updates all dynamic sections of the prompt for the next cycle. This includes refreshing the state description and available tools, appending the new output to the gathered information, and updating the last executed command section.

4 Implementation

We use Python 3.10 as our primary programming language. Docker is used to containerize and isolate command executions for enhanced reliability and reproducibility. RepairAgent is built on top of the AutoGPT framework and GPT-3.5-0125 from OpenAI. To parse and interact with Java code, we use ANTLR.

5 Findings/Results

To evaluate the approach, RepairAgent was tested to answer four main research questions regarding its effectiveness, costs, component importance, and tool usage.

5.1 Experimental Setup

- **Datasets:** Evaluated on all 835 bugs from the Defects4J [Just et al. 2014] dataset (v1.2 and v2.0). To assess generalization and mitigate data leakage risks, a random sample of 100 bugs from the newer GitBug-Java [Silva et al. 2024] dataset was also used.
- **Baselines:** Compared against three state-of-the-art iterative repair techniques: ChatRepair [Xia and Zhang 2023], ITER [Ye and Monperrus 2024], and SelfAPR [Ye et al. 2022b].
- **Metrics:** Evaluated based on *plausible* patches (passes all tests) and *correct* patches (syntactically matches or is semantically consistent with the developer’s fix).

5.2 Effectiveness (RQ1)

Evaluations performed across the Defects4J benchmark indicate that RepairAgent effectively resolved **164 functional bugs** (186 plausible), establishing capabilities beyond baseline tools like ChatRepair and ITER.

As shown in Table 3, RepairAgent particularly excels in Defects4Jv2, fixing 90 bugs compared to ChatRepair’s 48. Furthermore, it successfully fixed **39 unique bugs** that were not fixed by any of the prior state-of-the-art baselines (see Fig. 4).

Table 3. Results on Defects4J.

Project	Bugs	Plausible	Correct	ChatRepair	ITER
Chart	26	14	11	15	10
Closure	174	27	27	37	18
Math	106	29	29	32	0
Defects4Jv1.2	395	96	74	114	57
Defects4Jv2	440	90	90	48	—
Total	835	186	164	162	57

This performance is especially visible in multi-line fixes due to the contextual integration of search features. Table 4 shows RepairAgent outperforming baselines on complex, multi-line bugs. Further cross-validation against the GitBug-Java set shows functional adaptation across varied setups, successfully finding 13 correct fixes [Silva et al. 2024].

Table 4. Distribution of fixes by location type

Bug type	RepairAgent	ChatRepair	ITER	SelfAPR
Single-line	115	133	36	83
Multi-line*	46	29	14	24
Multi-file	3	0	4	3

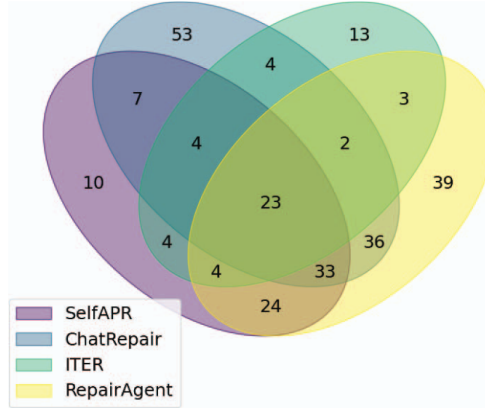


Fig. 4. Intersection of the fixes set with related work.

5.3 Costs of the Approach (RQ2)

Financial and temporal evaluations tracking token usage show an average financial overhead of approximately **14 cents per bug** resolution attempt (around 270k tokens) under standard GPT-3.5 pricing.

The median time taken to address a bug is 920 seconds (Fig. 5a). This is highly efficient compared to prior work; for instance, ITER reports a median bug fixing time of 4.57 hours per bug [Ye and Monperrus 2024].

5.4 Ablation Study (RQ3)

To understand the impact of different configurations, an ablation study was performed on a random set of 100 Defects4J bugs (Table 5).

- **No search tools:** Fixes drop by half, and costs double due to the agent reading long sequences of code instead of searching.
- **No state machine:** The agent loses structure, fixing fewer bugs and increasing costs.
- **Single-cycle memory:** Removing long-term memory causes a massive drop in effectiveness (only 6 correct fixes) as the agent repeats commands and forgets file names.
- **Realistic localization:** Using GZoltar [Campos et al. 2012] instead of perfect fault localization drops fixing capability by 25% and increases costs, as the agent spends extra time localizing the bug.

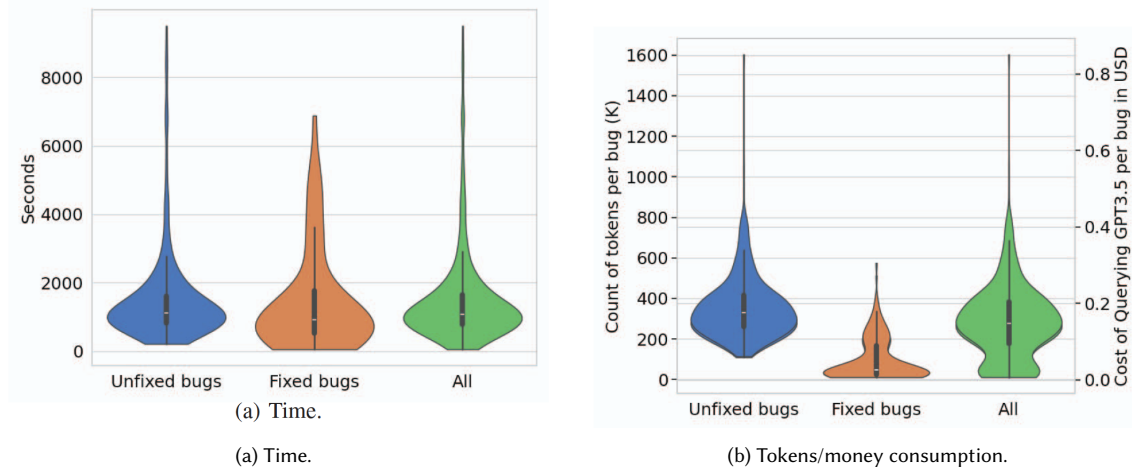


Fig. 5. Distribution of cost metrics per bug.

Table 5. Different configurations of RepairAgent.

Configuration	Plausible	Correct	Cost(\$)
No search tools	14	11	0.28
No state machine	18	14	0.31
Single-cycle memory	9	6	0.08
Realistic localization	16	16	0.29
RepairAgent (default)	23	21	0.16

5.5 Usage of Tools (RQ4)

On average, RepairAgent makes 35 tool invocations per bug. Fig. 6 shows that *write_fix* is the most frequently called tool. Interestingly, *run_tests* is used infrequently because the *write_fix* tool automatically invokes the test suite upon applying a patch.

6 Discussion

6.1 Qualitative Insights

Qualitative tracing indicates that access to structural tools yields better insights into functional anomalies. The agent’s ability to actively retrieve information—such as the code of failing tests, snippets retrieved via *find_similar_api_calls*, and feedback from applied fixes—allowed it to fix a new set of complex bugs without higher costs.

However, complex modifications sometimes introduce unintended changes to straightforward single-line defects. While RepairAgent clearly outperforms prior work on multi-line bugs, it occasionally fails to fix simpler, single-line bugs because the agent overcomplicates the solution. A potential remedy for future work is to initially limit the complexity of candidate fixes to nudge the agent toward simple solutions first.

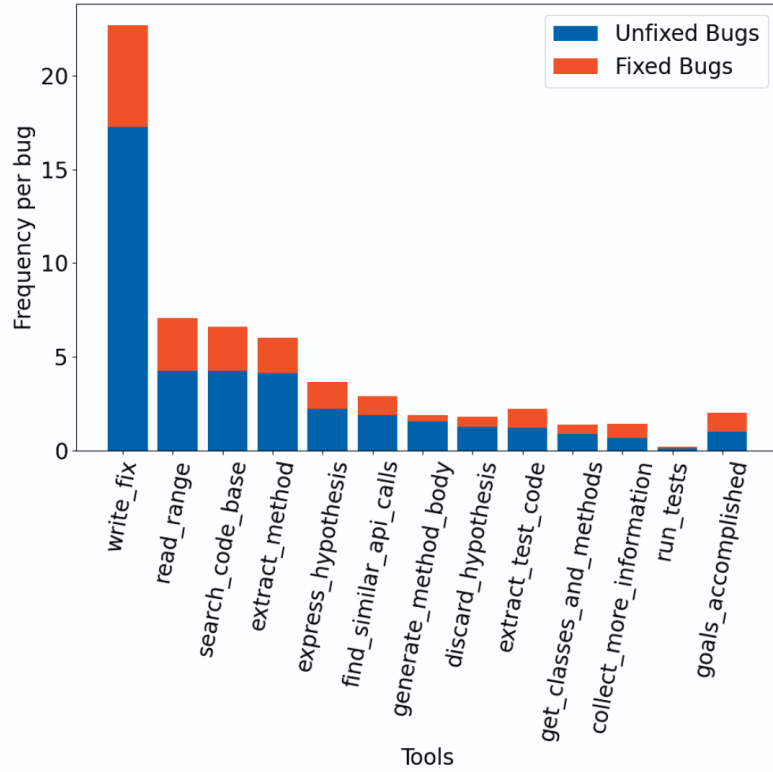


Fig. 6. Frequency of tool invocations (average per bug).

6.2 Threats to Validity and Limitations

- **Data Leakage:** Foundational models like GPT-3.5 may have seen parts of the Defects4J Java projects during pre-training. This risk is shared with competitors like ChatRepair [Xia and Zhang 2023], but is mitigated by our evaluation on the newer GitHub-Java dataset [Silva et al. 2024].
- **Missing Test Cases:** Defects4J guarantees at least one failing test case per bug, which may not reflect real-world scenarios where error-revealing tests are unavailable.
- **Fault Localization:** The default setup relies on perfect fault localization. As shown in the ablation study, relying on secondary spectrum coverage frameworks like GZoltar [Campos et al. 2012] introduces imprecision that can lead to suboptimal repair suggestions.
- **Non-deterministic Outputs:** The inherently non-deterministic nature of LLMs means outcomes may vary between consecutive runs.

7 Related Work

- **Non-learning-based program repair:** Automated program repair (APR) [Le Goues et al. 2019] has a rich history. Traditional approaches treat it as a search problem using manually designed mutation rules and fix patterns [Le et al. 2016; Le Goues et al. 2012; Liu et al. 2019]. Others derive rules from human-written patches

[Bader et al. 2019; Bavishi et al. 2019; Kim et al. 2013], utilize symbolic constraints [Ke et al. 2015; Mechtaev et al. 2016; Nguyen et al. 2013; Xuan et al. 2016], or integrate repair into static analysis [Jain et al. 2023; Liu et al. 2023; van Tonder and Le Goues 2018] and code transplantation [Yang et al. 2022]. APR has even seen industrial deployment [Bader et al. 2019; Marginean et al. 2019]. Beyond functional bugs, techniques exist for syntax errors [Gupta et al. 2019; Sakkas et al. 2022; Wang et al. 2018], performance bottlenecks [Yu and Pradel 2017], vulnerabilities [Harer et al. 2018], type errors [Chow et al. 2024], deep learning issues [Zhang et al. 2021], and build errors [Tarlow et al. 2020].

- **Learning-based program repair:** Early ML approaches focused on ranking candidate fixes [Long and Rinard 2016]. Recent work shifted to generation, using Neural Machine Translation (NMT) to map buggy code to fixed code [Chen et al. 2021b; Gupta et al. 2017; Lutellier et al. 2020; Tufano et al. 2019], predicting tree transformations [Li et al. 2020; Zhu et al. 2021], and employing repair-specific neural architectures and training regimes [Vasic et al. 2019; Ye et al. 2022b,a]. For a comprehensive overview, see [Zhang et al. 2023]. Unlike these, RepairAgent uses a general-purpose LLM rather than a task-specific model.
- **LLMs for APR and SE:** LLMs are increasingly applied to APR via one-time prompt interactions [Jiang et al. 2023; Joshi et al. 2023; Xia et al. 2023]. The current state-of-the-art relies on iterative feedback loops [Hidvégi et al. 2024; Kang et al. 2023a; Xia and Zhang 2023; Ye and Monperrus 2024]. Beyond repair, LLMs are used for code completion [Chen et al. 2021a; Shrivastava et al. 2023], fuzzing [Xia et al. 2024], unit testing [Alshahwan et al. 2024; Feng and Chen 2024; Kang et al. 2023b; Lemieux et al. 2023; Ryan et al. 2024; Schäfer et al. 2024], and multi-step editing [Bairi et al. 2023]. However, **none of these approaches are agent-based**. RepairAgent fundamentally differs by letting the model autonomously plan and execute actions.
- **LLM-based agents:** The concept of LLMs autonomously planning and performing complex tasks is relatively new [Wang et al. 2023]. RepairAgent is inspired by prior work on augmenting LLMs with API tools [Mialon et al. 2023; Patil et al. 2023; Schick et al. 2023] and code execution capabilities [Gao et al. 2022]. To the best of our knowledge, **RepairAgent is the first LLM-based agent applied to program repair**. Following the initial public release of this work, other SE agents have emerged, such as Copra for theorem proving [Thakur et al. 2024], AutoCodeRover [Zhang et al. 2024], and SWE-agent [Yang et al. 2024], highlighting the massive potential of this paradigm.

8 Conclusion

This paper presents a pioneering technique for bug repair based on an autonomous agent powered by Large Language Models (LLMs). By equipping the LLM with IDE-like tools and a dynamic prompt structure, we treat the model not as a simple text generator, but as an active participant in the debugging process. Through extensive experimentation on the Defects4J benchmark, we validate the effectiveness and financial viability of our approach. Further exploration and refinement of autonomous agent-based techniques—specifically by equipping them with the right tools—will help generalize automated repair to even more difficult and diverse types of software bugs.

Data Availability

Our code and data are available at: <https://github.com/sola-st/RepairAgent>

Alternatively: <https://doi.org/10.5281/zenodo.14470521>

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